

RayGlides 3W/4W EV Kit Cooling Controller - Assembly Notes

This document provides assembly line guidelines for solder paste application, components mounting, and visual inspection.

1. Moisture Sensitivity Level (MSL) Handling

- * **ESP32-S3-WROOM-1 Module**: Rated at **MSL 3**. If exposed to humidity for longer than 168 hours outside of protective vacuum packaging, it must be baked at 125°C for 24 hours prior to reflow soldering to prevent internal delamination (popcorning).
- * **LM5017MR Regulator**: Rated at **MSL 3**. Follow standard J-STD-033 guidelines for moisture barrier bag exposure limits.

2. Stencil Design and Solder Paste Printing

- * **Stencil Material**: Stainless steel, laser-cut, 0.12 mm (5 mils) thickness.
- * **LM5017 Exposed Pad Stencil**: The stencil opening under the LM5017 thermal pad must be segmented into a 2×2 grid (matrix array) with 0.5 mm bridges. This prevents component tilting or excessive solder paste squeezed out onto surrounding signal pins during reflow.
- * **Solder Paste Alignment**: Inspect paste printing registration prior to placing components. Solder coverage on pads must be $\geq 90\%$.

3. Visual Quality Inspection Criteria (IPC-A-610 Class 2)

1. **Solder Fillet**: Solder joints on SOT-23 (Q1), SOT-23-6 (ESD1), TSOT26 (U2), SOIC-8 (U6), and SO-PowerPAD-8 (U1) must show a positive concave meniscus fillet, showing good wetting.
2. **Thermal Vias**: Solder voiding inside the thermal vias under the LM5017 exposed pad must not exceed 25% of the total pad area (measured via X-ray inspection).
3. **Through-Hole Connectors**: Verify that Molex connector solder pins project 0.5 mm to 1.5 mm beyond the bottom solder pad, showing full barrel fill (minimum 75% vertical fill).